FPVspeed™

DIY Module Series

FPV Video Receiver

FS345Pro

3300～5110MHz -93±2dBm SPI + 128CH & RSSI

1.通用规格（General）

FS345Pro	
Weight	5.5g±0.1g
Dimensions	L37 x W26.3 x H4.5mm
DC Characteristics	
Power Supply	5.0V
Current Consumption	380mA ±30mA @5V
Environmental Specification	
Operating Temperature	-10~+65°C
Storing Temperature	-30~+85°C
Operating Humidity	85%RH
RF	
Receiving Frequency Range	3300~5110MHz
Voltage Standing Wave Ratio	2:1
Demodulation System	FM/PLL
IF	480MHz
ANT. Input Impedance	50Ω, Typ.
LO Frequency Stabilization	±200kHz
LO Frequency Precision	±200kHz
LO Control	PLL
Input LO Leak	-65dBm
Receiving Sensitivity	-93dBm±2dBm
Input Level Range	-93 dBm ~+5 dBm
Video Characteristics	
Video Output Impedance	75Ω, Typ.
Video Output Level	1±0.2Vp-p, Typ.
Video Polarity	Negative
Video Frequency Respone	±5dB, Max. 50Hz~6MHz
Differential Gain	±5%, Max
Differential Phase	±5Deg., Max
3Db IF WIDEBAND	16.5MHz
S/N	38dB, Min

Applications

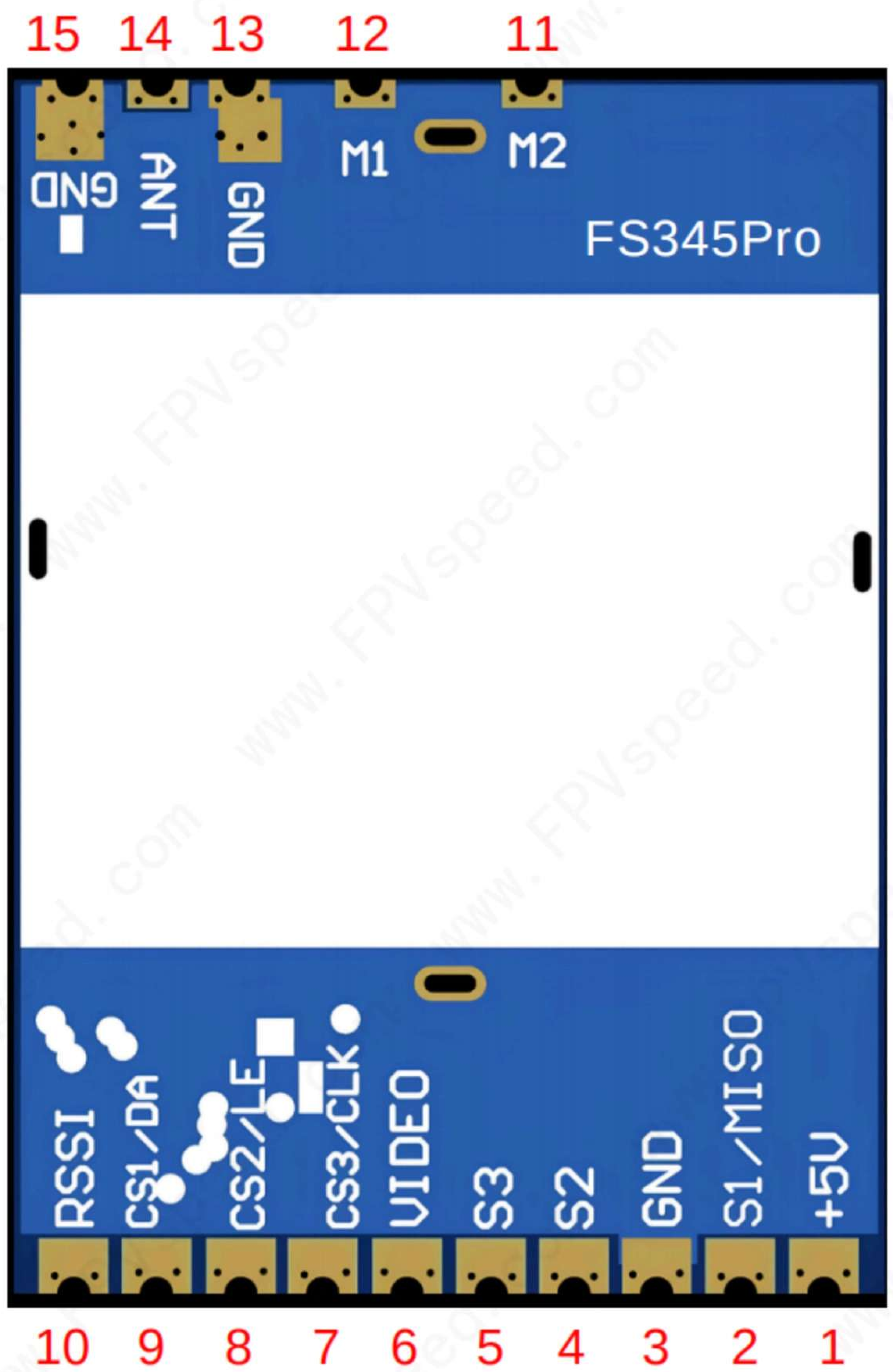
- FPV Racing Competition
- Wireless AV Transmission System
- Security Surveillance Communication
- Scientific Research and Education

2.测试条件（Test Condition）

Ambient temperature	25°C
Video input voltage	1.0Vp-p,150 (IRE)

3.引脚分布（Pin Assignment）

Pin No.	Function	
1	+5V IN	
2	S1, Frequency SEL.	
3	GND	
4	S2, Frequency SEL.	
5	S3, Frequency SEL.	
6	Video Out	
7	SPI_CLK / CS3	
8	SPI_LE / CS2	
9	SPI_DATA / CS1	
10	RSSI	
11	M2	1) M1=1, M2=1, SPI Mode 2) M1=0, M2=0, SPI Mode 3) M1=0, M2=1, A Switch Mode 4) M1=1, M2=0, B Switch Mode
12	M1	
13	GND	
14	ANT	
15	GND	



Mode Selection	M1 / M2	Logic Definition
SPI Mode	1 / 1	0 = GND / Low level 1 = Floating / High level
	0 / 0	
A Switch mode	0 / 1	
B Switch mode	1 / 0	

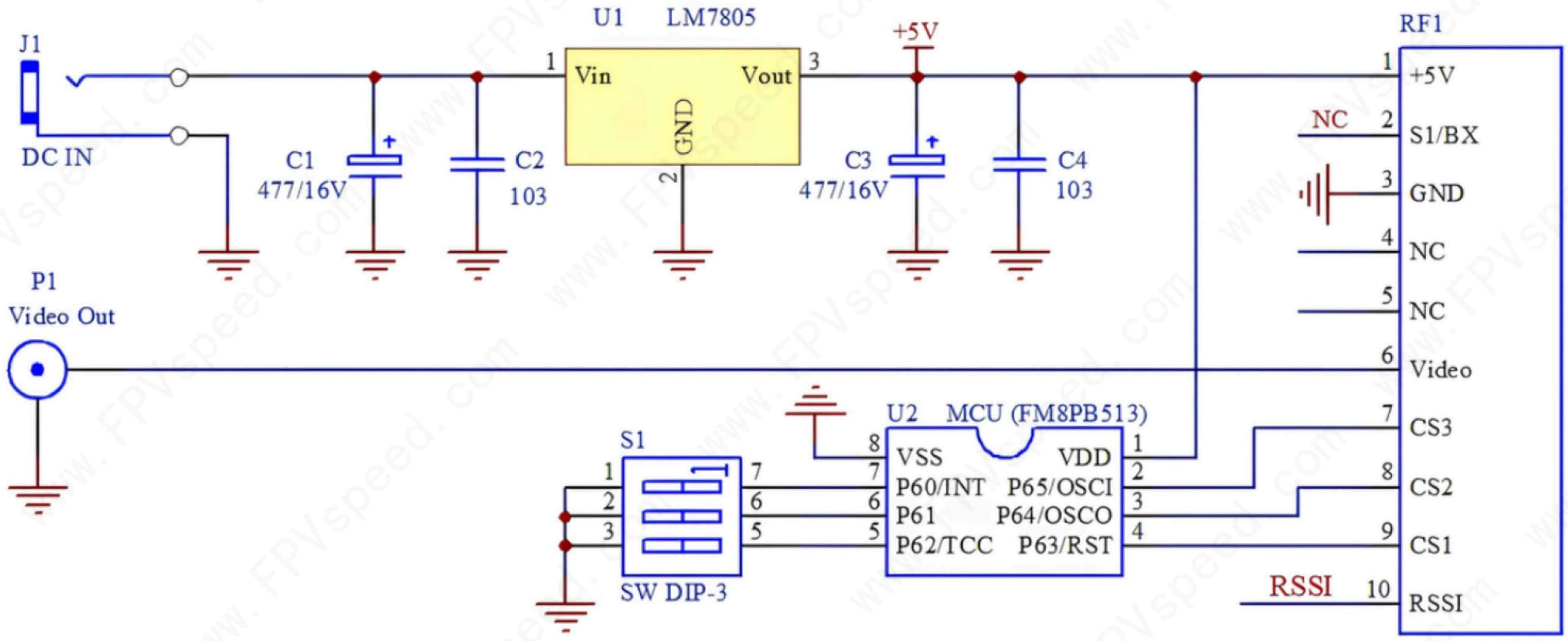
- The **FS345Pro** is a module that supports CVBS signal output. It features low latency and strong anti-interference capabilities, and is widely used in fields such as FPV (First-Person View) drone flight, outdoor entertainment, racing competitions, and professional aerial photography. It allows developers to conduct functional expansion and customized development based on the existing SPI protocol or GPIO switch.

SPI mode + GPIO switch mode 128CH
3300~5110 MHz

4.SPI控制（SPI Control Mode）

- When both **M1** and **M2** pins are left floating (high level) or pulled to GND (low level), the module operates in SPI mode, which allows frequency configuration via the SPI interface.
- the frequency range can be set to 3300~5110MHz, 1MHz steps.
- Maximum CLK frequency: 400 KHz (clock cycle = 2.5 μs)
- The high-level holding time of CLK shall not be less than 1 μs.
- The SPI configuration reference document [**SPIDemo.c**] can be downloaded from the official website FPVspeed.com or requested from the sales staff.

SPI Mode Application Circuit



5.开关控制及频道表（Switch Control & Channel Table）

- When pin **M1** is set to low level (GND) and pin **M2** is high level (floating), the module operates in **A** Switch mode.
- When pin **M2** is set to low level (GND) and pin **M1** is high level (floating), the module operates in **B** Switch mode.
- The preset frequency (one of the 128 fixed frequencies listed in the table below) is selected via the combined logic levels of S3, S2, S1, CS3, CS2 and CS1.

FS345Pro A Channel Table

A Switch Mode Channel Setting Table（MHz）										
Logic level definition: 0 = GND / Low level 1 = Floating / High level			CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
			CS3 / CS2 / CS1							
			0/0/0	0/0/1	0/1/0	0/1/1	1/0/0	1/0/1	1/1/0	1/1/1
FR 1	S3/S2/S1	0/0/0	3300	3320	3340	3360	3380	3400	3420	3440
FR 2		0/0/1	3460	3480	3500	3520	3540	3560	3580	3600
FR 3		0/1/0	3620	3640	3660	3680	3700	3720	3740	3760
FR 4		0/1/1	3780	3800	3820	3840	3860	3880	3900	3920
FR 5		1/0/0	3940	3960	3980	4000	4020	4040	4060	4080
FR 6		1/0/1	4100	4120	4140	4160	4180	4200	4220	4240
FR 7		1/1/0	4260	4280	4300	4320	4340	4360	4380	4400
FR 8		1/1/1	4420	4440	4460	4480	4500	4520	4540	4560
M1 = 0 ， M2 = 1										

FS345Pro B Channel Table

B Switch Mode Channel Setting Table（MHz）										
Logic level definition: 0 = GND / Low level 1 = Floating / High level			CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
			CS3 / CS2 / CS1							
			0/0/0	0/0/1	0/1/0	0/1/1	1/0/0	1/0/1	1/1/0	1/1/1
FR 1	S3/S2/S1	0/0/0	3900	3925	3950	3975	4000	4025	4050	4075
FR 2		0/0/1	4100	4125	4150	4175	4200	4225	4250	4275
FR 3		0/1/0	4300	4325	4350	4375	4400	4425	4450	4475
FR 4		0/1/1	4500	4525	4550	4575	4600	4625	4650	4675
FR 5		1/0/0	4700	4725	4750	4775	4800	4825	4850	4875
FR 6		1/0/1	4900	4925	4950	4975	5000	5025	5050	5075
FR 7		1/1/0	4510	4535	4560	4585	4610	4635	4660	4685
FR 8		1/1/1	4900	4930	4960	4990	5020	5050	5080	5110
M1 = 1 ， M2 = 0										

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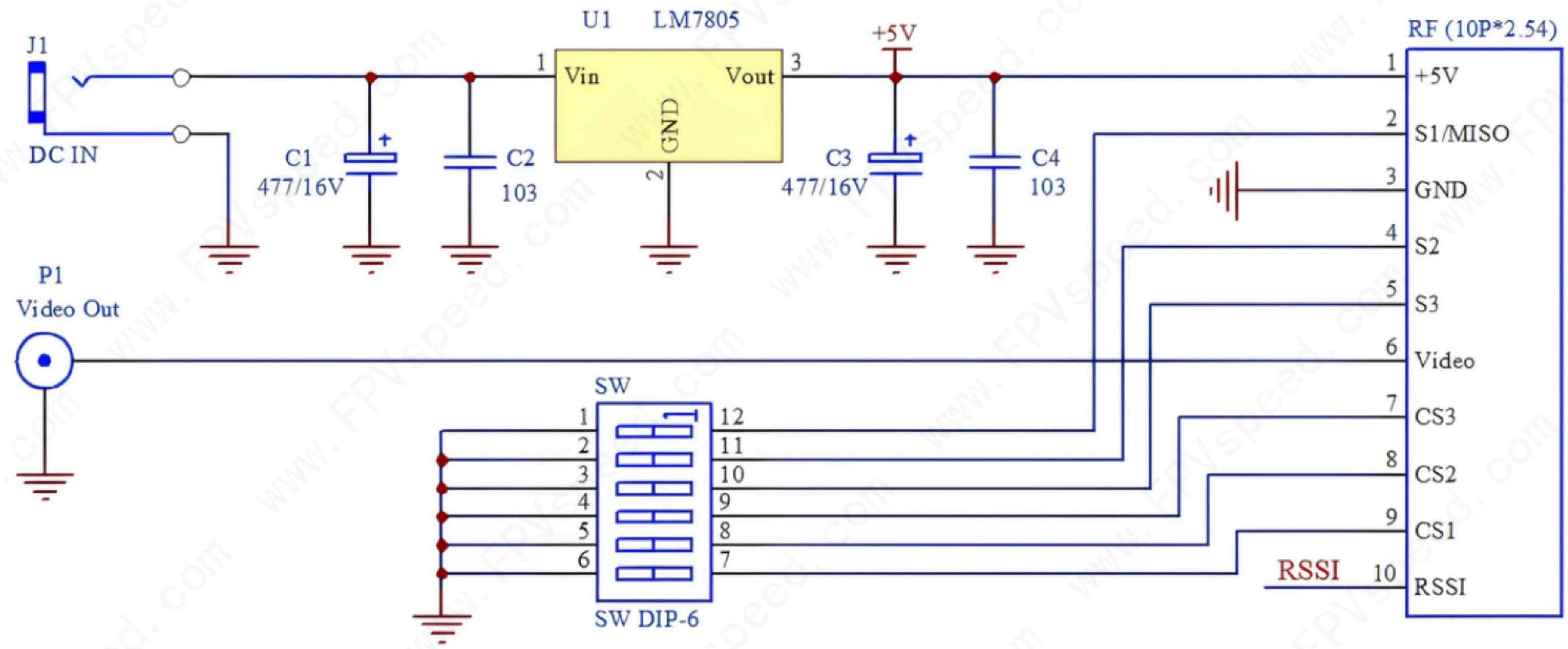
Switch Mode Application Circuit



6.结构尺寸（Dimension）